

PRACTICAL NO 11

AIM: Reshaping data using pivot_longer() and pivot_wider() (R).

OUTPUT:

The first screenshot shows the initial data loading and the first pivot operation. The R script in the Source pane includes comments and code to read a CSV file, create a PatientID, and pivot the data from wide to long format. The Environment pane on the right shows the objects created: df, df_risk, Diabetes_Prediction, long_df, risk_pivot, and wide_df.

```
> # R Script: Reshaping data with pivot_longer() and pivot_wider()
> # Dataset: Diabetes Prediction data
>
> library(dplyr)
> library(tidyr)
>
> # 1. SETUP: Create and Import Data
>
> # Read data and add a PatientID (essential for tracking rows during pivots)
> df <- read.csv("Diabetes_Prediction.csv", na.strings = c("", "NA")) %>%
+ mutate(
+   PatientID = row_number(), # Row-wise unique ID
+   Risk_Probability = diagnosed_diabetes, # Original probability
+   Risk_Percent = diagnosed_diabetes * 100 # Same value in percent
+ ) %>%
+ # selecting fewer columns for clarity (similar to sir's code)
+ select(PatientID, id, Risk_Probability, Risk_Percent)
>
> print("--- 1. original wide Data (Diabetes) ---")
[1] "--- 1. original wide Data (Diabetes) ---"
> print(head(df))
  PatientID id Risk_Probability Risk_Percent
1      100000 0.5069355      50.69355
2      200001 0.7197535      71.97535
3      300002 0.7763152      77.63152
4      400003 0.4276024      42.76024
5      500004 0.9088413      90.88413
6      600005 0.6323532      63.23532
>
> # 2. PIVOT_LONGER (wide to Long)
>
> # Scenario:
> # You want to combine 'Risk_Probability' and 'Risk_Percent' into a single column
> # called 'value', with a label column called 'Metric'.
> # Useful for: Plotting multiple metrics on one graph (e.g., ggplot2).
>
> long_df <- df %>%
+ pivot_longer(
```

The second screenshot shows the continuation of the R script, including the second pivot operation to reshape the data back to wide format. The R script includes comments and code to pivot the data from long to wide format. The Environment pane on the right shows the objects created: df, df_risk, Diabetes_Prediction, long_df, risk_pivot, and wide_df.

```
> pivot_longer(
+   cols = c(Risk_Probability, Risk_Percent), # The columns we want to stack
+   names_to = "Metric", # Name for the new column containing header names
+   values_to = "value" # Name for the new column containing the numbers
+ )
>
> print("--- 2. Long Format (pivot_longer) ---")
[1] "--- 2. Long Format (pivot_longer) ---"
> # Notice how PatientID 1 now appears TWICE (once for Risk_Probability, once for Risk_Percent)
> print(head(long_df, 6))
# A tibble: 6 x 4
  PatientID id Metric value
  <int> <int> <chr> <dbl>
1      100000 0 Risk_Probability 0.507
2      100000 0 Risk_Percent 50.7
3      200001 0 Risk_Probability 0.720
4      200001 0 Risk_Percent 72.0
5      300002 0 Risk_Probability 0.776
6      300002 0 Risk_Percent 77.6
>
> # 3. PIVOT_WIDER (Long to wide)
>
> # Scenario:
> # You have the 'long_df' created above, and you want to spread it back out
> # so that every Metric gets its own column again.
> # Useful for: Creating report tables or preparing data for machine learning.
>
> wide_df <- long_df %>%
+ pivot_wider(
+   names_from = Metric, # which column contains the new header names?
+   values_from = value # which column contains the data to fill cells?
+ )
>
> print("--- 3. wide Format (back to wide) ---")
[1] "--- 3. wide Format (back to wide) ---"
> print(head(wide_df))
# A tibble: 6 x 4
```

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DATA ANALYSIS WITH SAS / SPSS / R

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Source

```
R - R 4.5.2 - ~/ABHISHEK DINESH SINGH/
> # 3. PIVOT_WIDER (Long to wide)
> # =====
> # Scenario:
> # You have the 'long_df' created above, and you want to spread it back out
> # so that every metric gets its own column again,
> # Useful for: Creating report tables or preparing data for machine learning.
> wide_df <- long_df %>%
+ pivot_wider(
+   names_from = Metric, # which column contains the new header names?
+   values_from = value # which column contains the data to fill cells?
+ )
>
> print("--- 3. wide Format (Back to wide) ---")
[1] "--- 3. wide Format (Back to wide) ---"
> print(head(wide_df))
# A tibble: 6 x 4
  PatientID id Risk_Probability Risk_Percent
  <int> <dbl> <dbl> <dbl>
1 1 700000 0.507 50.7
2 2 700001 0.720 72.0
3 3 700002 0.776 77.6
4 4 700003 0.428 42.8
5 5 700004 0.909 90.9
6 6 700005 0.632 63.2
>
> # =====
> # 4. ADVANCED EXAMPLE (Reshaping for Reporting)
> # =====
> # Here we create risk categories (Low / Medium / High) based on Risk_Probability
> # and then spread them into columns.
> # Similar idea to sir ke category A/B/C wale example.
> df_risk <- df %>%
+ mutate(
+   RiskGroup = case_when(
+     Risk_Probability < 0.33 ~ "Low",
+     Risk_Probability < 0.66 ~ "Medium",
+     TRUE ~ "High"
+   )
+ )
>
> # =====
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+   )
+ )
>
> risk_pivot <- df_risk %>%
+ select(PatientID, RiskGroup, Risk_Percent) %>%
+ pivot_wider(
+   names_from = RiskGroup, # columns = Low / Medium / High
+   values_from = Risk_Percent # Cell values = Risk percentage
+ )
>
> print("--- 4. Risk Group Pivot (Spreading Risk Categories) ---")
[1] "--- 4. Risk Group Pivot (Spreading Risk Categories) ---"
> print(head(risk_pivot))
# A tibble: 6 x 4
  PatientID Medium High Low
  <int> <dbl> <dbl> <dbl> <dbl>
1 1 50.7 NA NA
2 2 NA 72.0 NA
3 3 NA 77.6 NA
4 4 42.8 NA NA
5 5 NA 90.9 NA
6 6 63.2 NA NA
> library(dplyr)
```

Environment

History

Connections

Tutorial

Data

df	300000 obs. of 4 variables	
df_risk	300000 obs. of 5 variables	
diabetes_Prediction	300000 obs. of 2 variables	
long_df	600000 obs. of 4 variables	
risk_pivot	300000 obs. of 4 variables	
wide_df	300000 obs. of 4 variables	

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ABHISHEK DINESH SINGH.Rproj	218 B	Dec 8, 2025, 11:22 AM
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Source

```
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+     Risk_Probability < 0.66 ~ "Medium",
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+ )
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> risk_pivot <- df_risk %>%
+ select(PatientID, RiskGroup, Risk_Percent) %>%
+ pivot_wider(
+   names_from = RiskGroup, # columns = Low / Medium / High
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+ )
>
> print("--- 4. Risk Group Pivot (Spreading Risk Categories) ---")
[1] "--- 4. Risk Group Pivot (Spreading Risk Categories) ---"
> print(head(risk_pivot))
# A tibble: 6 x 4
  PatientID Medium High Low
  <int> <dbl> <dbl> <dbl> <dbl>
1 1 50.7 NA NA
2 2 NA 72.0 NA
3 3 NA 77.6 NA
4 4 42.8 NA NA
5 5 NA 90.9 NA
6 6 63.2 NA NA
> library(dplyr)
```

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